

Giridhera Ramanan S

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Portfolio: giridhera-ramanan.github.io | github.com/giridhera-ramanan

Profile Summary

AI & Data Science graduate and Full-Stack Developer with hands-on experience building AI-powered web applications using Next.js, FastAPI, PostgreSQL, and modern LLM APIs. Developed and deployed production-style applications including RAG chatbots, computer vision systems, and full-stack SaaS projects. Passionate about backend engineering, AI systems, and building scalable software with clean architecture.

Education

Panimalar Engineering College Chennai City Campus, B.Tech Artificial Intelligence & Data Science (CGPA: 8.22/10)	Nov 2022 – May 2026
Sitadevi Garodia Hindu Vidyalaya Hr. Sec. Matriculation, Higher Secondary School Education (Percentage: 87.3%)	May 2022
Sitadevi Garodia Hindu Vidyalaya Hr. Sec. Matriculation, Secondary School Education (Percentage: 69.7%)	Mar 2020

Experience

Java Development Trainee, <i>Robowaves - IN</i> - Solved 100+ Java programming exercises covering arrays, strings, collections, object-oriented programming, and exception handling. - Implemented reusable Java programs emphasizing clean coding practices and modular design. - Strengthened problem-solving and debugging skills through practical coding assignments and hands-on exercises.	June 2025 – Aug 2025
IoT Intern, <i>KUBE RESEARCH LABS PRIVATE LIMITED - Chennai, IN</i> - Developed and integrated BLE-based RSSI tracking and ultrasonic sensor logic with Arduino Uno for an autonomous smart cart, achieving reliable real-time object avoidance within a 30cm detection range. - Designed and executed a 50+ test validation protocol for autonomous cart prototypes, reducing hardware fault rate and driving measurable improvements in navigation stability and sensor response time.	Nov 2024 – Jan 2025
AI/ML Intern, <i>InternPe - Remote</i> - Developed five machine learning projects including Breast Cancer Detection, Diabetes Prediction, Email Spam Detection, Car Price Prediction, and IPL Match Prediction using Scikit-learn. - Performed data preprocessing, feature engineering, model training, and evaluation using Python, Pandas, and Scikit-learn. - Built reusable prediction pipelines and documented experiments for model comparison and testing.	Aug 2024 – Sep 2024

Skills

Programming Languages: Python • Java • JavaScript • TypeScript • SQL
Frontend: React.js • Next.js • HTML5 • CSS3 • Tailwind CSS • Vite
Integration & Backend: REST API design • FastAPI • Node.js • JWT Authentication • third-party & LLM API integration • Webhook-style provider fallback routing
Data & Cloud: PostgreSQL (Neon) • MySQL • SQLite • AWS • Render • Vercel • Netlify
AI / Agentic: LangChain • Retrieval-Augmented Generation (RAG) • Prompt Engineering • NLP • TensorFlow • PyTorch • Scikit-learn • OpenCV • NumPy • Pandas
Security & Reliability: PII masking • Encrypted storage • Rate limiting • Request-level monitoring & latency benchmarking
Developer Tools: Git • GitHub • Linux • Postman • Jupyter Notebook • VS Code

Projects

Aura – AI-Powered Mental Health Journal <i>Next.js • FastAPI • PostgreSQL (Neon) • Multi-LLM APIs • Vercel</i> - Led a 4-person team as final-year capstone lead, owning backend and API integration; locked scope through an upfront ideation phase, split work into scheduled units, and delivered ahead of deadline with a full unit-testing pass. - Built a full-stack AI journaling platform for mood tracking and guided self-reflection; integrated 3 LLM providers with automatic fallback routing, sustaining 100% request completion across 55 production calls (40 - Secured journal entries with JWT authentication, encrypted storage, regex-based PII masking, and per-route rate limiting, achieving zero PII exposure across all endpoints. - Deployed frontend and backend using Netlify and Vercel while configuring PostgreSQL on Neon for cloud-based persistence.	Live Demo: aura-journal.netlify.app
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Portfolio AI Chatbot

Live Demo: giridhera-ramanan.github.io

FastAPI • LangChain • Groq Llama 3.1 • RAG

- Built an AI recruiter assistant embedded on my portfolio, letting visitors query my projects and experience in real time instead of reading a static page; used long-context RAG inference in place of an external vector database, achieving 0.65s average response time (0.43s best case).
- Embedded a responsive JavaScript chat widget into the personal portfolio with real-time API communication.

Road Accident Detection (RAD)

Github: github.com/giridhera-ramanan/RAD

TensorFlow • OpenCV • SMOTE • MobileNetV2

- Built a real-time video system detecting road accidents from camera footage for faster emergency response; tuned the detection threshold for a safety-first 68% recall (vs. 40% precision) across a held-out 2,000-image set, running at 104–147 img/s batch throughput.
- Trained the model using the "Car Crash or Collision Prediction Dataset" from Kaggle (~10,000 labeled images), applying preprocessing, augmentation, and class balancing before model training.

Posture Monitor – Real-Time Posture Correction System

Github: github.com/giridhera-ramanan/posture-monitor

Python • OpenCV • MediaPipe (Pose Landmarker) • Computer Vision

- Built a real-time posture monitoring application using MediaPipe Pose Landmarker with personalized multi-stage calibration for different users.
- Designed adaptive thresholds that combine the measured distance between a user's good and bad posture with a noise floor derived from their own calibration jitter (std-dev), preventing false positives across different webcams and body types.
- Engineered a sustained-drift alert pipeline (8s hold time, 15s cooldown) with desktop + audio notifications via plyer and a terminal-beep fallback, plus a startup self-test so notification failures surface immediately instead of mid-session.
- Added a headless background mode and session analytics (% time in good/bad posture, alert count), and made all file I/O resolve relative to the script's own path to eliminate environment-dependent path bugs.

Moodflix – Movie Discovery App

Live Demo: moodflix04.netlify.app

React 19 • Vite • Tailwind CSS • Appwrite • TMDB API

- Developed a responsive movie discovery platform using React, Vite, Tailwind CSS, and the TMDB API.
- Integrated Appwrite to store search history and generate trending movie recommendations.
- Implemented debounced searching, lazy loading, and graceful error handling to improve performance and user experience.
- Designed a mobile-first responsive interface supporting desktop, tablet, and mobile devices.

Technical Highlights

- Built and deployed 6+ full-stack and AI applications.
- Integrated multiple commercial LLM providers including Groq, Together AI, and OpenRouter.
- Developed REST APIs using FastAPI with PostgreSQL backend.
- Deployed applications on Render, Vercel, and Netlify.
- Experienced with computer vision, NLP, Retrieval-Augmented Generation (RAG), and cloud deployment.

Certifications

Java Programming for Beginners - Simplilearn & SkillUp

AWS Academy Cloud Foundations – AWS Academy Graduate

Fundamentals of Machine Learning & Deep Learning – NPTEL Online Certification

Artificial Intelligence for Real-World Applications – TCS iON

Journey to Cloud: Envisioning Your Solution – IBM SkillsBuild

Introduction to Cybersecurity – Cisco